BASE CHANGE OF A K-FLAT DIFFERENTIAL GRADED ALGEBRA COMPUTES THE DERIVED RESTRICTION AND COMMUTES WITH THE GROUND-RING DERIVED TENSOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the canonical comparison map c_M, the claim for property-(P) modules from which the three parts follow through the derived-tensor definitions, the propagation of isomorphism and K-flatness through sums of shifts, the induction over the split filtration with more-algebra-lemma-K-flat-two-out-of-three, and the filtered-colimit conclusion are all retained verbatim.

Editorial note. The label lemma-base-change-K-flat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Derived tensor products through property (P) resolutions. Same chapter, Lemma lemma-derived-bc; More on Algebra, Section more-algebra-section-derived-tensor-product. Fixes the definition of the derived tensor functors, so that the claim about property-(P) modules converts into the three stated parts by the defining formalism.
- K-flatness bookkeeping for sums, extensions and colimits. More on Algebra, Lemmas more-algebra-lemma-colimit-K-flat and more-algebra-lemma-K-flat-two-out-of-three. Propagate K-flatness and the isomorphism c from the algebra through sums of shifts, the split filtration extensions, and the final filtered colimit.

Source reference. Original reference 09LS: sources/stacks-dga/pinned-original-source.tex, source lines 5541–5541. Prerequisite pointer invoked at proof line(s) 5757; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06Y2: sources/stacks-chow/tags.txt, source lines 8871–8871. Prerequisite pointer invoked at proof line(s) 5775; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06Y3: sources/stacks-chow/tags.txt, source lines 8872–8872. Prerequisite pointer invoked at proof line(s) 5765; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 06XY: sources/stacks-chow/tags.txt, source lines 8867–8867. Prerequisite pointer invoked at proof line(s) 5758; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $R \to R'$ be a ring map. Let (A, d) be a differential graded R-algebra. Let (A', d) be the base change, i.e., $A' = A \otimes_R R'$. If A is K-flat as a complex of R-modules, then*

(1) *$- \otimes^{\mathbf{L}}_A A' : D(A, d) \to D(A', d)$ is equal to the right derived functor of*
$$K(A, d) \longrightarrow K(A', d), \quad M \longmapsto M \otimes_R R'$$
(2) *the diagram*
$$\begin{array}{ccc} D(A, d) & \xrightarrow[-\otimes^{\mathbf{L}}_A A']{} & D(A', d) \\ {\scriptstyle restriction}\downarrow & & \downarrow{\scriptstyle restriction} \\ D(R) & \xrightarrow[-\otimes^{\mathbf{L}}_R R']{} & D(R') \end{array}$$
commutes, and
(3) *if M is K-flat as a complex of R-modules, then the differential graded A'-module $M \otimes_R R'$ represents $M \otimes^{\mathbf{L}}_A A'$.*

Proof. For any differential graded A-module M there is a canonical map
$$c_M : M \otimes_R R' \longrightarrow M \otimes_A A'$$
Let P be a differential graded A-module with property (P). We claim that c_P is an isomorphism and that P is K-flat as a complex of R-modules. This will prove all the results stated in the lemma by formal arguments using the definition of derived tensor product in Lemma 09LS and More on Algebra, Section 06XY.

Let $F_\bullet$ be the filtration on P showing that P has property (P). Note that c_A is an isomorphism and A is K-flat as a complex of R-modules by assumption. Hence the same is true for direct sums of shifts of A (you can use More on Algebra, Lemma 06Y3 to deal with direct sums if you like). Hence this holds for the complexes $F_{p+1}P/F_pP$. Since the short exact sequences
$$0 \to F_pP \to F_{p+1}P \to F_{p+1}P/F_pP \to 0$$
are split exact as sequences of graded modules, we can argue by induction that c_{F_pP} is an isomorphism for all p and that F_pP is K-flat as a complex of R-modules (use More on Algebra, Lemma 06Y2). Finally, using that $P = \operatorname{colim} F_pP$ we conclude that c_P is an isomorphism and that P is K-flat as a complex of R-modules (use More on Algebra, Lemma 06Y3). $\square$